FIT1047 Weekly Reflection Template (for references only - you can use your own structure)

Word count: at least 100 words (per week)

Week 1

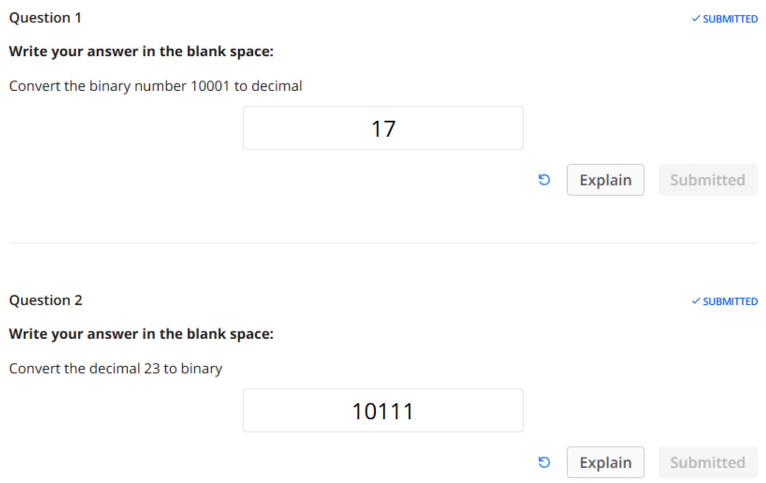
Key Learnings:

- Number representation in binary is interesting as it shows how our computers store and interpret binary into readable numbers for the users. The conversion between binary, hexadecimal and decimal is also useful as they all have their own use cases. ASCII is the representation of text and other symbols on our computers using a 8-bit system, or one byte, which can only represent 256 characters. The AND, OR, NOT gate allowed me to learn how transistors are used in our computer's CPU to allow for calculations and different processes.

Personal Insights:

- While I have learned about the binary systems before, learning it formally here in Monash gave me a different perspective on the matter entirely. I now fully understand how a computer operates under binary, and how this system is the basis for logic gates and all the computing in the future.

Application of Knowledge:



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- I was able to apply my knowledge on conversion between binary to decimal and vice versa when doing the “Test Your Knowledge” in the Ed Forum. Using the multiplication method.
- One application for this is when doing low level programming, like programming in C, where I will need to deal with pointers and memory addresses.

Challenges Faced:

- Some challenges that I faced during the first week were accommodating to the amount of new information I had to absorb before the first week of classes started. It was overwhelming at first, but in the end I managed to get all the pre-reading completed before class through effective time management.

Future Goals:

- Moving forward, I plan to write more notes on the pre-reading materials before entering the classes of the week, as I realised that it is actually quite difficult to keep up with the class without prior knowledge by reading materials on Ed.

Week 2

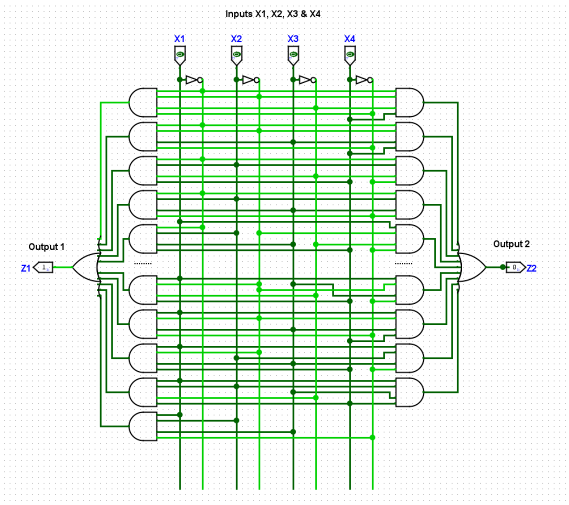
Key Learnings:

- My key takeaway from this week was learning boolean logic, and logic gates. These two topics taught me the fundamentals of how a Central Processing Unit (CPU) is built on, which are basically transistors / logic gates. Moreover, learning how to form boolean functions and equations, by using truth tables and using karnaugh maps. Using the karnaugh maps, I learned how to simplify the boolean equations using the Sum Of Products (SOP), Product Of Sum (POS) and boolean laws. These methods allow for simplification of the boolean expressions which will save logic gates needed in a particular circuit.

Personal Insights:

- Boolean logic and logic gates are all quite new to me, I found it extremely intriguing as I now am able to understand the basic building blocks of how a computer processes a command or instruction. I learned that the logic gates in the CPU are represented by transistors. Learning the truth tables and knowing how to simplify them using Karnaugh maps made me realize that no matter how complex a boolean operation is, it can still be broken down into a simpler form. Overall, this chapter allowed me to have a better understanding and more confidence when dealing with computers.

Application of Knowledge:

- 

- Using the knowledge of K-Maps and truth tables to form a boolean expression, I was able to apply it when making the actual circuit in the Logisim software. Knowing when to use the NOT, AND, and OR gates were crucial in making the entire circuit work.

Challenges Faced:

- The Karnaugh map was a particularly challenging concept for me to understand at first. I did not understand the rules to the grouping and how the K-Map itself was formed. However, after consulting with my friends on the matter, I was fortunate to be guided by them and learned the right methods to approach the K-Map.

Future Goals:

- Moving on to next week, I hope to keep researching and dive deeper into the topic by reading the pre-class materials on Ed Forum.

Week 3

Key Learnings:

- I learned the Con Neumann architecture, which is a classic computer design model that a lot of modern computers follow and use. Moreover, I was introduced to machine code and assembly language, where I learned that machine code is the binary instructions that a CPU understands, which depends on their Instruction Set Architecture (ISA), such as x86 or ARM. MARIE was used for us to learn and understand how it works.Furthermore, I learned about adders and multiplexers, and how these components work together to form the arithmetic logic unit (ALU). Lastly, we were taught how memory works in computers, and how it can remember things using a loop in logic gates. This is particularly important and insightful for me as I now truly understand how a computer can store memory.

Personal Insights:

- Learning how the logic gates are used to make memory, to make adders and ultimately ALU, is the most important part for me in this chapter. Prior to this chapter, I was merely just using the computers under the assumption that everything just works. But now, I have a much clearer understanding of the inner workings and the immense engineering put into each of the devices that we take for granted everyday.

Application of Knowledge:

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LOAD A
ADD B
Output
Halt
A, DEC 2
B, DEC 3
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- Using MARIE code, I was able to make some simple programs from almost the lowest level of programming.

Challenges Faced:

- The challenges I faced was understanding the details of how the digital circuits are made into the Decoder and the Multiplexer. While I understand that it is not necessary for me to know 100% how each component is formed, I still find it interesting to learn about it. To solve this, I researched outside of the materials provided in Ed to further clarify my understanding.

Future Goals:

- After this week, I will continue to practice MARIE code to further understand the logic gates and the construction of the adders, memories and flip flop circuits. As I understand that these will probably be coming up in the future assessments.